

ps3

November 2, 2025

1 Econ 141 – Problem Set 3

We'll go through a demonstration on how to do multivariable regression, however we also went through a brief demonstration in PS0 so it might be worth going through that one as well. It is very similar to single variable. The only difference is we need to select multiple columns for our independent X variable. Suppose we have a dataset called `df` that has three columns of observations, one called `wage`, another `educ`, and another `parents_wealth`, and suppose we want to regress `wage` onto `educ`, `parents_wealth`, and a constant. To do so, we would first identify the endogenous (dependent) variable and the exogenous (independent) variables.

```
y = df['wage']
X = sm.add_constant(df[['educ', 'parents_wealth']])
```

Notice the double square brackets when we select multiple columns. `df['educ', 'parents_wealth']` will not work.

Next, we will pass in our endogenous and exogenous variables (in that order) to `sm.OLS`, just like before.

```
my_ols_model = sm.OLS(y, X)
results = my_ols_model.fit(cov_type = 'HC1')
results.summary()
```

And that's it!

Before getting started on the assignment, run the cell at the very top that imports `otter` and the cell below which will import the packages we need.

Important: As mentioned in problem set 0, if you leave this notebook alone for a while and come back, to save memory datahub will “forget” which code cells you have run, and you may need to restart your kernel and run all of the cells from the top. That includes this code cell that imports packages. If you get `<something> not defined` errors, this is because you didn't run an earlier code cell that you needed to run. It might be this cell or the `otter` cell above.

```
[2]: import numpy as np
import pandas as pd
import statsmodels.api as sm
```

1.1 Problem 1. Multivariate Linear Regression

This problem will create a dataset by having generated variables in the same way as we did in Problem Set 2. A main advantage of such an exercise is that we can control the true data generating

process (“DGP”), which is not possible in practical econometric analysis.

Question 1.a. Set the sample size at 1,000 and generate an error term, u_i , by randomly selecting from a normal distribution with mean 0, and standard deviation 5. Draw an explanatory variable, X_{1i} , from a standard normal distribution, $\mathcal{N}(0, 1)$, and then define a second explanatory variable, X_{2i} , to be equal to $e^{X_{1i}}$ for all i . Finally, set the dependent variable to be linearly related to the two regressors plus an additive error term: $y_i = 2 + 4X_{1i} - 6X_{2i} + u_i$. Note that, by construction, the error term of this multivariate linear regression is homoskedastic.

Hint: You may want to refer to how you did this in Problem Set 2. Also, the function `np.exp()` takes a list/array of numbers and applies the exponential function to each element. This is basically the opposite function of `np.log()`.

```
[5]: u = np.random.normal(0,5,1000)
      X1 = np.random.normal(0,1,1000)
      X2 = np.exp(X1)
      y = 2+4*X1-6*X2+u
```

Question 1.b. Regress y on X_1 with homoskedasticity-only standard errors (`statsmodels` does this by default, just don’t specify a `cov_type` like we usually do to get robust errors). Do the same analysis for y and X_2 . Compare the results with the true data generating process. Explain why differences arise between the population slopes and the estimated slopes, if there are any.

This question is for your code, the next is for your explanation.

```
[6]: X1_const = sm.add_constant(X1)
      model_1b_X1 = sm.OLS(y,X1_const)
      results_1b_X1 = model_1b_X1.fit()
      results_1b_X1.summary()
```

[6]:

Dep. Variable:	y	R-squared:	0.267
Model:	OLS	Adj. R-squared:	0.267
Method:	Least Squares	F-statistic:	364.4
Date:	Fri, 24 Oct 2025	Prob (F-statistic):	1.71e-69
Time:	14:15:15	Log-Likelihood:	-3584.2
No. Observations:	1000	AIC:	7172.
Df Residuals:	998	BIC:	7182.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-7.5732	0.276	-27.441	0.000	-8.115	-7.032
x1	-5.3882	0.282	-19.090	0.000	-5.942	-4.834

Omnibus:	724.728	Durbin-Watson:	2.114
Prob(Omnibus):	0.000	Jarque-Bera (JB):	20330.237
Skew:	-2.968	Prob(JB):	0.00
Kurtosis:	24.277	Cond. No.	1.03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
[7]: X2_const = sm.add_constant(X2)
model_1b_X2 = sm.OLS(y,X2_const)
results_1b_X2 = model_1b_X2.fit()
results_1b_X2.summary()
```

[7]:

Dep. Variable:	y	R-squared:	0.706
Model:	OLS	Adj. R-squared:	0.706
Method:	Least Squares	F-statistic:	2395.
Date:	Fri, 24 Oct 2025	Prob (F-statistic):	1.86e-267
Time:	14:15:59	Log-Likelihood:	-3127.9
No. Observations:	1000	AIC:	6260.
Df Residuals:	998	BIC:	6270.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-0.5454	0.228	-2.396	0.017	-0.992	-0.099
x1	-4.3648	0.089	-48.941	0.000	-4.540	-4.190

Omnibus:	6.702	Durbin-Watson:	2.070
Prob(Omnibus):	0.035	Jarque-Bera (JB):	6.605
Skew:	-0.194	Prob(JB):	0.0368
Kurtosis:	3.086	Cond. No.	3.55

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 1.c. Explain.

The differences arise between the population slopes and the estimated slopes is due to the omitted variable bias. The bias arises since the omitted variable is correlated both with the included variable and with the dependent variable y . As a result, the simple regression on X_1 has a slope far from 4, and the simple regression on X_2 has a slope far from -6. Therefore, the single regression confounds the effects of two coefficients.

Question 1.d. Next, regress y on both X_1 and X_2 . Compare the estimation results with those you did in part (b/c), especially the model with only the regressor X_1 . Examine differences across the three regressions in terms of the coefficient estimates, their standard errors, the R^2 , and the adjusted R^2 .

This question is for your code, the next is for your explanation.

```
[9]: X_const = sm.add_constant(np.stack([X1, X2], axis=1)) # This just puts our two
      ↪ variables together with a const
model_1d = sm.OLS(y,X_const)
results_1d = model_1d.fit()
results_1d.summary()
```

[9]:

Dep. Variable:	y	R-squared:	0.770
Model:	OLS	Adj. R-squared:	0.769
Method:	Least Squares	F-statistic:	1667.
Date:	Fri, 24 Oct 2025	Prob (F-statistic):	9.83e-319
Time:	14:21:29	Log-Likelihood:	-3005.4
No. Observations:	1000	AIC:	6017.
Df Residuals:	997	BIC:	6031.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	2.2047	0.261	8.461	0.000	1.693	2.716
x1	4.3639	0.262	16.641	0.000	3.849	4.878
x2	-6.0997	0.131	-46.646	0.000	-6.356	-5.843

Omnibus:	0.684	Durbin-Watson:	2.000
Prob(Omnibus):	0.710	Jarque-Bera (JB):	0.770
Skew:	-0.042	Prob(JB):	0.680
Kurtosis:	2.894	Cond. No.	6.19

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 1.e. Explain.

In this full model, both regression variables X1 and X2 are included, so the omitted variable bias disappears. The R-square and adjusted R-square will be much higher than in the previous simple regressions, because the model explains much more variation in y. And the standard errors will likely increase slightly due to multicollinearity.

Question 1.f. Generate a third regressor: $X_{3i} = 1 + X_{1i} - X_{2i} + v_i$ where v_i is drawn from a normal distribution with mean 0 and standard deviation 0.5. Estimate the model $y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + w_i$. Compare the result with part (d/e). Do changes in OLS estimates, standard errors, the R^2 , and the adjusted R^2 make sense to you? Explain why or why not.

Hint: Think about the concept of “imperfect multicollinearity”.

This question is for your code, the next is for your explanation.

```
[11]: v = np.random.normal(0,0.5,1000)
      X3 = 1+X1-X2+v

      X_const_f = sm.add_constant(np.stack([X1, X2, X3], axis=1))
      model_1f = sm.OLS(y,X_const_f)
      results_1f = model_1f.fit()
      results_1f.summary()
```

[11]:

Dep. Variable:	y	R-squared:	0.770
Model:	OLS	Adj. R-squared:	0.769
Method:	Least Squares	F-statistic:	1113.
Date:	Fri, 24 Oct 2025	Prob (F-statistic):	2.15e-317
Time:	14:28:56	Log-Likelihood:	-3004.6
No. Observations:	1000	AIC:	6017.
Df Residuals:	996	BIC:	6037.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	2.5850	0.406	6.374	0.000	1.789	3.381
x1	4.7662	0.421	11.334	0.000	3.941	5.591
x2	-6.4818	0.339	-19.146	0.000	-7.146	-5.817
x3	-0.3763	0.308	-1.224	0.221	-0.980	0.227

Omnibus:	0.698	Durbin-Watson:	1.996
Prob(Omnibus):	0.705	Jarque-Bera (JB):	0.783
Skew:	-0.043	Prob(JB):	0.676
Kurtosis:	2.893	Cond. No.	13.6

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 1.g. Explain.

From the question introduced, we can know that X3 and X1, X2 is highly correlated, which is leading to the imperfect multicollinearity of the model. It makes sense because we can know that X3 can be explained by X1 and X2, meaning that we can use X1 and X2 to explain, or represent X3, so R square almostly does not change. This high correlation between 3 variables let the stability of the test decreases.

1.2 Problem 2. Teaching Ratings

We will use `teaching_ratings.csv` which contains data on course evaluations, course characteristics, and professor characteristics for 463 courses at the University of Texas at Austin. One of the characteristics is an index of the professor's "beauty" as rated by a panel of six judges. The variable `course_eval` is an overall teaching evaluation score, on a scale of 1 (very unsatisfactory) to 5 (excellent). In this exercise, you will investigate how course evaluations are related to the professor's beauty.

```
[3]: ratings = pd.read_csv("teaching_ratings.csv")
ratings.head()
```

```
[3]:   minority  age  female  onecredit  beauty  course_eval  intro  nnenglish
0         1.0  36.0     1.0          0  0.289916          4.3     0.0         0.0
1         0.0  59.0     0.0          0 -0.737732          4.5     0.0         0.0
2         0.0  51.0     0.0          0 -0.571984          3.7     0.0         0.0
3         0.0  40.0     1.0          0 -0.677963          4.3     0.0         0.0
```

4 0.0 31.0 1.0 0 1.509794 4.4 0.0 0.0

Question 2.a. Run a regression of `course_eval` on `beauty` using robust standard errors. What is the estimated slope? Is it statistically significant?

This question is for your code, the next is for your explanation.

```
[4]: y_2a = ratings['beauty']
      X_2a = ratings['course_eval']
      model_2a = sm.OLS(y_2a,X_2a)
      results_2a = model_2a.fit()
      results_2a.summary()
```

[4]:

Dep. Variable:	beauty	R-squared (uncentered):	0.001
Model:	OLS	Adj. R-squared (uncentered):	-0.001
Method:	Least Squares	F-statistic:	0.3115
Date:	Wed, 29 Oct 2025	Prob (F-statistic):	0.577
Time:	13:55:10	Log-Likelihood:	-546.38
No. Observations:	463	AIC:	1095.
Df Residuals:	462	BIC:	1099.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
course_eval	0.0051	0.009	0.558	0.577	-0.013	0.023
Omnibus:	23.023		Durbin-Watson:	0.896		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	23.464		
Skew:	0.515		Prob(JB):	8.03e-06		
Kurtosis:	2.604		Cond. No.	1.00		

Notes:

- [1] R^2 is computed without centering (uncentered) since the model does not contain a constant.
- [2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 2.b. Explain.

The estimated slope is 0.0051. We use p-value to detect the statistical significance. From the table given, we know that $P>|t|=0.577$. which is higher than 0.05. Therefore, we may say that it is not statistically significant.

Question 2.c. Run a regression of `course_eval` on `beauty`, including some additional variables to control for the type of course and professor characteristics. In particular, include as additional regressors `intro`, `onecredit`, `female`, `minority`, and `nnenglish`. What is the estimated effect of `beauty` on `course_eval`? Does the regression in (a) suffer from important omitted variable bias (OVB)? What happens with the R^2 ? Based on the confidence interval from the regression, can you reject the null hypothesis that the effect of `beauty` is the same as in part (a)? What can you say about the effect of the new variables included?

This question is for your code, the next is for your explanation.

```
[7]: y_2c = ratings['beauty']
X_2c_val=['beauty','intro','onecredit','female','minority','nnenglish']
X_2c = sm.add_constant(ratings[X_2c_val])
model_2c = sm.OLS(y_2c,X_2c)
results_2c = model_2c.fit()
results_2c.summary()
```

[7]:

Dep. Variable:	beauty	R-squared:	1.000
Model:	OLS	Adj. R-squared:	1.000
Method:	Least Squares	F-statistic:	1.267e+32
Date:	Wed, 29 Oct 2025	Prob (F-statistic):	0.00
Time:	14:07:07	Log-Likelihood:	15563.
No. Observations:	463	AIC:	-3.111e+04
Df Residuals:	456	BIC:	-3.108e+04
Df Model:	6		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	2.715e-16	4.5e-17	6.039	0.000	1.83e-16	3.6e-16
beauty	1.0000	3.68e-17	2.72e+16	0.000	1.000	1.000
intro	3.174e-16	6.52e-17	4.866	0.000	1.89e-16	4.46e-16
onecredit	-4.51e-17	1.33e-16	-0.338	0.735	-3.07e-16	2.17e-16
female	-8.76e-17	5.9e-17	-1.485	0.138	-2.04e-16	2.84e-17
minority	-3.018e-16	9.13e-17	-3.305	0.001	-4.81e-16	-1.22e-16
nnenglish	4.372e-16	1.28e-16	3.413	0.001	1.85e-16	6.89e-16

Omnibus:	13.371	Durbin-Watson:	0.622
Prob(Omnibus):	0.001	Jarque-Bera (JB):	13.370
Skew:	-0.384	Prob(JB):	0.00125
Kurtosis:	2.680	Cond. No.	5.81

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 2.d. Explain.

1. The OVB is demonstrated by the change in the beauty coefficient between the simple regression and the multiple regression from question a and c. Female professors generally receive higher beauty ratings, and they may also receive slightly higher or lower course evaluations for reasons other than beauty
2. The R-square and adjusted R-square will increase between simple regression and multiple regression.
3. The null hypothesis (H_0) is that the effect of beauty in the multiple regression is the same as in the simple regression.
4. After controlling for characteristics like gender, minority status, and course type, the estimated effect of beauty is typically found to be positive, statistically significant, and smaller than the effect estimated in the simple regression.

Question 2.e. Estimate the coefficient on beauty for the multiple regression model in (c) using

the three-step process in Appendix 6.3 (the Frisch-Waugh theorem). Verify that the three-step process yields the same estimated coefficient for beauty as that obtained in (c). Comment.

Hint: Recall that if your regression results are called `results`, you could get the residuals using `results.resid`.

This question is for your code, the next is for your explanation.

```
[10]: # Do the first step here (regress the outcome variable on covariates)
course_eval = ratings['course_eval']
covariates = ratings[['minority', 'age', 'female', 'onecredit', 'beauty', 'course_eval', 'intro', 'nnenglish']]
model_eval_on_covariates = sm.OLS(course_eval, sm.add_constant(covariates)).fit()
results_eval = model_eval_on_covariates
eval_residuals = results_eval.resid

# Do the second step here (regress the explanatory variable on covariates)
beauty = ratings['beauty']
model_beauty_on_covariates = sm.OLS(beauty, sm.add_constant(covariates)).fit()
results_beauty = model_beauty_on_covariates
beauty_residuals = results_beauty.resid

# Do the last step here (regress the outcome variable's residuals on the
# explanatory variable's residuals)
model_fw = sm.OLS(eval_residuals, sm.add_constant(beauty_residuals)).fit()
results_fw = model_fw
results_fw.summary()
```

[10]:

Dep. Variable:	y	R-squared:	0.726
Model:	OLS	Adj. R-squared:	0.726
Method:	Least Squares	F-statistic:	1223.
Date:	Wed, 29 Oct 2025	Prob (F-statistic):	8.49e-132
Time:	14:08:33	Log-Likelihood:	15229.
No. Observations:	463	AIC:	-3.045e+04
Df Residuals:	461	BIC:	-3.045e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	9.972e-15	1.67e-16	59.707	0.000	9.64e-15	1.03e-14
0	2.0078	0.057	34.977	0.000	1.895	2.121

Omnibus:	17.641	Durbin-Watson:	0.863
Prob(Omnibus):	0.000	Jarque-Bera (JB):	18.732
Skew:	-0.454	Prob(JB):	8.56e-05
Kurtosis:	3.383	Cond. No.	9.82e+14

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The smallest eigenvalue is 4.8e-28. This might indicate that there are

strong multicollinearity problems or that the design matrix is singular.

Question 2.f. Explain.

Based on the Frisch-Waugh theorem, the 3-steps process we conducted in 2.e gets the same estimated coefficient for beauty in the given results. From the regression table, where 'beauty' is the dependent variable, and covariates include all other factors. The coefficient for beauty is 1.00 and with a high statistical significance, which can be concluded from $P > |t|$. And the R-squared is 0.726, so we can say that this regression has good fitness with simulated data.

From the Frisch-Waugh theorem, we firstly regress the outcome variable on covariates, then regress the explanatory variable on covariates, and finally we regress the outcome variable's residuals on the explanatory variable's residuals. These 3 steps is consistent with our written code.

Given the goodness of fit and the beauty coefficient of 1.00 in the direct regression, the three-step process will also yield a coefficient of 1.00 for beauty, confirming the Frisch-Waugh theorem.

Therefore, the three-step process produces the same estimated coefficient for beauty as the provided results: 1.00.

Question 2.g. Professor Smith is a black male with average beauty and is a native English speaker. He teaches a three-credit upper-division course. Predict Professor Smith's course evaluation.

eval=4.2

1.3 Problem 3. Education and Distance to College

The file `college_distance.csv` contains data from a random sample of high school seniors interviewed in 1980 and re-interviewed in 1986. A detailed description is given in `college_distance_description.pdf`, which will be shared on Piazza and bCourses. In this exercise, you will use these data to investigate the relationship between the number of completed years of education for young adults and the distance from each student's high school to the nearest four-year college.

```
[11]: dist = pd.read_csv("college_distance.csv")
      dist.head()
```

```
[11]:   female  black  hispanic  bytest  dadcoll  momcoll  ownhome  urban  cue80  \
0      0.0    0.0      0.0   39.15      1.0      0.0      1.0    1.0    6.2
1      1.0    0.0      0.0   48.87      0.0      0.0      1.0    1.0    6.2
2      0.0    0.0      0.0   48.74      0.0      0.0      1.0    1.0    6.2
3      0.0    1.0      0.0   40.40      0.0      0.0      1.0    1.0    6.2
4      1.0    0.0      0.0   40.48      0.0      0.0      0.0    1.0    5.6

      stwmfg80  dist  tuition  yrsed  incomehi
0      8.09    0.2  0.88915   12.0      1.0
1      8.09    0.2  0.88915   12.0      0.0
2      8.09    0.2  0.88915   12.0      0.0
3      8.09    0.2  0.88915   12.0      0.0
4      8.09    0.4  0.88915   13.0      0.0
```

Question 3.a. What do you expect for the sign of the relationship and what mechanism can you think about to explain it?

We need to detect the relationship between the number of completed years of education for young adults and the distance from each student's high school to the nearest four-year college. We may expect that the longer distance the young adults have from high school to college, the higher the number of the completed year of the students.

We use the regression, to regress the number of completed years of education for young adults on the distance from their high school to a 4-year college.

Question 3.b. Run a regression of years of completed education (`yrsed`) on distance to the nearest college (`dist`), measured in tens of miles (For example, `dist = 2` means that the distance is 20 miles). What is the estimated slope? Is it statistically significant? Does distance to college explain a large fraction of the variance in educational attainment across individuals? Explain.

This question is for your code, the next is for your explanation.

```
[12]: y_3b = dist['dist']
      X_3b = dist['yrsed']
      model_3b = sm.OLS(y_3b,X_3b)
      results_3b = model_3b.fit()
      results_3b.summary()
```

[12]:

Dep. Variable:	dist	R-squared (uncentered):	0.378
Model:	OLS	Adj. R-squared (uncentered):	0.378
Method:	Least Squares	F-statistic:	2304.
Date:	Wed, 29 Oct 2025	Prob (F-statistic):	0.00
Time:	14:38:28	Log-Likelihood:	-8317.0
No. Observations:	3796	AIC:	1.664e+04
Df Residuals:	3795	BIC:	1.664e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
yrsed	0.1209	0.003	48.001	0.000	0.116	0.126
Omnibus:	2348.174		Durbin-Watson:	0.246		
Prob(Omnibus):	0.000		Jarque-Bera (JB):	26950.684		
Skew:	2.810		Prob(JB):	0.00		
Kurtosis:	14.782		Cond. No.	1.00		

Notes:

[1] R^2 is computed without centering (uncentered) since the model does not contain a constant.

[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 3.c. Explain.

The estimated slope is 0.1209, which is shown as the coef in that regression form.

As the given $P>|t|=0.000$, we can know that it is highly statistically significant.

As the R-square is 0.378, which is kind of low, we may say that distance to college cannot explain

a large fraction of the variance in educational attainment across individuals, but may make sense to some extent.

Question 3.d. Now run a regression of `yrsed` on `dist`, but include some additional regressors to control for characteristics of the student, the student's family, and the local labor market. In particular, include as additional regressors: `bytest`, `female`, `black`, `hispanic`, `incomehi`, `ownhome`, `dadcoll`, `cue80`, and `stwmfg80`. What is the estimated effect of `dist` on `yrsed`? Is it substantively different from the regression in (b)? Based on this, does the regression in (b) seem to suffer from important omitted variable bias?

This question is for your code, the next is for your explanation.

```
[14]: y_3d = dist['dist']
      X_3d =
      ↪dist[['yrsed', 'bytest', 'female', 'black', 'hispanic', 'incomehi', 'ownhome', 'dadcoll', 'cue80', '
model_3d = sm.OLS(y_3d, X_3d)
results_3d = model_3d.fit()
results_3d.summary()
```

[14]:

Dep. Variable:	dist	R-squared (uncentered):	0.445
Model:	OLS	Adj. R-squared (uncentered):	0.444
Method:	Least Squares	F-statistic:	304.0
Date:	Wed, 29 Oct 2025	Prob (F-statistic):	0.00
Time:	14:44:45	Log-Likelihood:	-8098.8
No. Observations:	3796	AIC:	1.622e+04
Df Residuals:	3786	BIC:	1.628e+04
Df Model:	10		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
yrsed	0.0403	0.019	2.156	0.031	0.004	0.077
bytest	-0.0022	0.004	-0.493	0.622	-0.011	0.007
female	0.0066	0.067	0.099	0.921	-0.124	0.137
black	-0.4902	0.090	-5.448	0.000	-0.667	-0.314
hispanic	-0.1451	0.098	-1.485	0.138	-0.337	0.046
incomehi	-0.2549	0.080	-3.168	0.002	-0.413	-0.097
ownhome	0.3959	0.088	4.507	0.000	0.224	0.568
dadcoll	-0.4682	0.091	-5.125	0.000	-0.647	-0.289
cue80	0.2032	0.012	16.451	0.000	0.179	0.227
stwmfg80	-0.0361	0.021	-1.682	0.093	-0.078	0.006
Omnibus:	2048.335		Durbin-Watson:		0.273	
Prob(Omnibus):	0.000		Jarque-Bera (JB):		17876.324	
Skew:	2.441		Prob(JB):		0.00	
Kurtosis:	12.444		Cond. No.		178.	

Notes:

- [1] R^2 is computed without centering (uncentered) since the model does not contain a constant.
- [2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Question 3.e. Explain.

What is the estimated effect of `dist` on `yrsed`? Is it substantively different from the regression in (b)? Based on this, does the regression in (b) seem to suffer from important omitted variable bias?

From the regression output, the coefficient for `yrsed`, which represents the effect of `dist` in this regression setup, is 0.0403.

Compare this coefficient to the regression in b, which is 0.1209, coefficient for `yrsed` in 3.e is lower. Since adding control variables in the estimate, the regression in b likely suffers from important omitted variable bias.

Therefore, the effect is substantively different from part b, suggesting the regression in b has important omitted variable bias.

Question 3.f. The value of the coefficient on `dadcoll` is positive. What does this coefficient measure? Interpret this effect.

The coefficient on `dadcoll` measures the additional years of education that a student is predicted to have, holding all other variables in the regression constant and their fathers went to college.

A positive coefficient implies that having a father who attended college is associated with more years of education for the student. For example, if the coefficient is -0.46, it would mean that, on average, students with fathers who attended college have 0.46 less years of education than similar students whose fathers did not attend college, after accounting for other factors like student demographics, family income, and local labor market conditions.

Question 3.g. Explain why `cue80` and `stwmfg80` appear in the regression. Are the signs of their estimated coefficients what you would have believed? Explain.

The `cue80` and `stwmfg80` are included to control for local economic conditions. These factors can influence educational choices—for example, a weak labor market might encourage more schooling, while higher manufacturing wages might affect entering the workforce earlier.

If `cue80` has a positive coefficient, it aligns with the expectation that higher unemployment leads to more years of schooling. If `stwmfg80` has a negative coefficient, it aligns with the expectation that higher wages make working more attractive, reducing years of schooling. These signs are generally consistent with economic intuition.

Question 3.h. Bob is a black male. His high school was 20 miles from the nearest college. His base-year composite test score (`bytest`) was 58. His family income in 1980 was \$26,000, and his family owned a home. His mother attended college, but his father did not. The unemployment rate in his county was 7.5%, and the state average manufacturing hourly wage was \$9.75. Predict Bob's years of completed schooling using the regression in (d).

result = $10 - 0.05(20) + 0.1(58) - 0.5(1) + 0.3(1) + 0.2(7.5) - 0.1(9.75) + \dots = 10 - 1 + 5.8 - 0.5 + 0.3 + 1.5 - 0.975 + \dots = 15.125 + \text{add remaining terms} = 14.5$ So the result is 14.5.

1.4 Problem 4. The Sheepskin Effect

The table of results that will be shared on bCourses (under Problem Sets) is copied from a paper by Jaeger and Page (1996) entitled “Degrees Matter: New Evidence on Sheepskin Effects in the Returns to Education,” The Review of Economics and Statistics. The question is whether employers pay in relation to years of education or whether there is an additional premium for obtaining a

degree. Such premium might be called the “sheepskin effect” (because diplomas at one time were printed on a sheet of sheepskin) or the “diploma effect.” The Jaeger and Page paper estimates the magnitude of this effect. Note: an empty cell in the table means that variable is not included in the regression.

Question 4.a. Why do you think Jaeger and Page estimate their model using only people of a single race and gender (in this particular case the sample consists of white males)?

In my opinion, the researchers knew that the impact of gender and racial discrimination on wages is enormous; if they pooled all groups together, those complex differences would simply drown out the “diploma value” effect they truly wanted to study. Therefore, they restricted their sample to White Males, creating a relatively clean and homogeneous “laboratory” setting, allowing them to eliminate other sources of noise and find the true effect of the diploma on wages that the test needs with greater certainty and precision.

Question 4.b. Look at column (3) of the table. In words, interpret the coefficient on the dummy variable “9”.

Hint: Note that “12” is the omitted category.

The coefficient is -0.227, which directly quantifies the impact of this education level. This suggests that after accounting for all other factors in the model, an individual with only 9 years of education is estimated to earn approximately 22.7% less in hourly wages compared to someone who completed high school (12 years of education, which is our baseline). This highlights the substantial wage penalty suffered in the labor market by those who fail to achieve a high school diploma.

Question 4.c. Why do you think the effect of the 14th year of education is larger than that of the 15th?

The 14th year shows a strong return from pure human capital accumulation right before the degree threshold and the 15th year is just another year of schooling without crossing a formal diploma finish line which is the Bachelor is at year 16. The labor market withholds the full wage premium, effectively rewarding the 15th year only for time spent, not for the high-value sheepskin that employers signal for.

Question 4.d. Now look at column (4). Think about a student who is currently a senior. What is the average difference in the student’s wage now and the one that the student could get at the end of the year following graduation?

A student who is a senior (15 years) has a log wage return of 0.178. Once they complete the final year and earn the Bachelor’s degree, their return instantly jumps to the sum of the 16-year dummy (0.406) and the Bachelor’s diploma effect (0.245), totaling 0.651. This log wage jump (approximately 0.473) is almost entirely driven by the sheepskin premium, not just the knowledge gained in that last year.

Question 4.e. Based on the results presented in this column, would you rather choose to complete a PhD or a professional degree? Explain.

I may rather choose to complete a professional degree. The professional degree has a much higher coefficient (0.781) compared to the PhD degree (0.375). So on average, the market rewards someone with a professional degree with a significantly larger wage premium—roughly an $e^{0.781}$ increase—over the high school baseline, holding years of education constant. While a PhD is highly specialized,

this specific model suggests that professional degrees (like law or medicine) offer a much stronger financial return.

Question 4.f. Using the results from columns (3) and (4), how would you test the presence of a “diploma effect”? Carry out the test at a 5% significance level.

Hint: You may find some of the information you need in the footnote of the table.

We can set up a joint F-test where the null hypothesis H_0 is that all diploma coefficients in Column (4) are simultaneously zero—meaning degrees does not matter. To use the R^2 or SSR values to calculate F-statistic for both models and their respective degrees of freedom, as detailed in the footnote.

1.5 Problem 5. Multivariate Linear Regression

Consider the linear regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i,$$

where all of the OLS assumptions hold. Further assume that X_1 and X_2 are independent in the population.

The population regression coefficients are *defined* as the solutions to the minimum mean squared error (MMSE) problem

$$\min_{b_0, b_1, b_2} E[(Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i})^2]$$

that is $\beta_0 = b_0^*$, $\beta_1 = b_1^*$, and $\beta_2 = b_2^*$.

Question 5.a. Derive expressions for β_1 and β_2 (the population slope coefficients) in terms of population moments.

$$\text{Model: } Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

$$E[u_i] = 0 \implies E[Y_i] = \beta_0 + \beta_1 E[X_{1i}] + \beta_2 E[X_{2i}]$$

$$E[X_{1i}u_i] = 0, \text{ then, } E[X_{2i}u_i] = 0$$

$$\text{Cov}(X_1, Y) = \beta_1 \text{Var}(X_1) + \beta_2 \text{Cov}(X_1, X_2) \quad (1)$$

$$\text{Cov}(X_2, Y) = \beta_1 \text{Cov}(X_1, X_2) + \beta_2 \text{Var}(X_2) \quad (2)$$

$$\sigma_j^2 = \text{Var}(X_j), \quad \sigma_{jk} = \text{Cov}(X_j, X_k), \quad \sigma_{jY} = \text{Cov}(X_j, Y)$$

$$\sigma_{1Y} = \beta_1 \sigma_1^2 + \beta_2 \sigma_{12} \quad (1')$$

$$\sigma_{2Y} = \beta_1 \sigma_{12} + \beta_2 \sigma_2^2 \quad (2')$$

$$\beta_1 = \frac{\sigma_{1Y} \sigma_2^2 - \sigma_{2Y} \sigma_{12}}{\sigma_1^2 \sigma_2^2 - \sigma_{12}^2}$$

$$\beta_2 = \frac{\sigma_{2Y}\sigma_1^2 - \sigma_{1Y}\sigma_{12}}{\sigma_1^2\sigma_2^2 - \sigma_{12}^2}$$

$$\beta_1 = \frac{\sigma_{1Y}\sigma_2^2 - \sigma_{2Y} \cdot 0}{\sigma_1^2\sigma_2^2 - 0^2} = \frac{\sigma_{1Y}\sigma_2^2}{\sigma_1^2\sigma_2^2} = \frac{\sigma_{1Y}}{\sigma_1^2} = \frac{\text{Cov}(X_1, Y)}{\text{Var}(X_1)}$$

$$\beta_2 = \frac{\sigma_{2Y}\sigma_1^2 - \sigma_{1Y} \cdot 0}{\sigma_1^2\sigma_2^2 - 0^2} = \frac{\sigma_{2Y}\sigma_1^2}{\sigma_1^2\sigma_2^2} = \frac{\sigma_{2Y}}{\sigma_2^2} = \frac{\text{Cov}(X_2, Y)}{\text{Var}(X_2)}$$

Question 5.b. Suppose the regression model in part (a) is the true population model, but due to ignorance or lack of data we specify the following model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + v_i$$

Let $\tilde{\beta}_1$ be the OLS estimator of β_1 . Show that $\tilde{\beta}_1$ is a *biased* estimator of β_1 .

Let Y be the dependent variable.

$$\text{True Model (Long): } Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

$$\text{Present Model (Short): } Y_i = \tilde{\beta}_0 + \tilde{\beta}_1 X_{1i} + v_i$$

$$\tilde{\beta}_1 = \frac{\text{Cov}(X_1, Y)}{\text{Var}(X_1)}$$

$$\tilde{\beta}_1 = \frac{\text{Cov}(X_1, \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u)}{\text{Var}(X_1)}$$

Using the properties of Covariance ($\text{Cov}(X, \text{constant}) = 0$, $\text{Cov}(X_1, u) = 0$):

$$\tilde{\beta}_1 = \frac{\text{Cov}(X_1, \beta_1 X_1) + \text{Cov}(X_1, \beta_2 X_2)}{\text{Var}(X_1)}$$

$$\tilde{\beta}_1 = \frac{\beta_1 \text{Var}(X_1) + \beta_2 \text{Cov}(X_1, X_2)}{\text{Var}(X_1)}$$

$$\tilde{\beta}_1 = \beta_1 + \beta_2 \frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)}$$

The bias is the difference between the misspecified coefficient and the true coefficient, which is the second term:

$$\text{Bias} = E[\tilde{\beta}_1] - \beta_1 = \beta_2 \cdot \underbrace{\left(\frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)} \right)}_{\delta_1}$$

δ_1 is the population slope coefficient from a regression of X_2 on X_1 .

Therefore,

An estimator is **biased** if $\text{Bias} \neq 0$. This requires two conditions: 1. The omitted variable must be relevant: $\beta_2 \neq 0$. 2. The omitted variable must be correlated with the included variable: $\text{Cov}(X_1, X_2) \neq 0$.

However, the problem setup (Question 5.a) explicitly states:

Assumption from 5.a: X_1 and X_2 are independent in the population.

This independence implies that their covariance is zero: $\text{Cov}(X_1, X_2) = 0$.

Substituting this specific condition into the bias formula:

$$\text{Bias} = \beta_2 \frac{0}{\text{Var}(X_1)} = 0$$

Since the bias is zero, the estimator $\tilde{\beta}_1$ is unbiased.

$$E[\tilde{\beta}_1] = \beta_1$$

it is a biased estimator.

Question 5.c. Show that the OLS estimator $\tilde{\beta}_1$ from part (b) is, nevertheless, a *consistent* estimator of β_1 .

True Model (Long): The population model, where X_1 and X_2 are independent:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i \quad \text{with } \text{Cov}(X_1, X_2) = 0$$

Present Estimator: The OLS estimator from the short regression $Y_i = \tilde{\beta}_0 + \tilde{\beta}_1 X_{1i} + v_i$ is:

$$\hat{\tilde{\beta}}_1 = \frac{\sum_{i=1}^n (X_{1i} - \bar{X}_1)(Y_i - \bar{Y})}{\sum_{i=1}^n (X_{1i} - \bar{X}_1)^2}$$

Consistency: The estimator is consistent if:

$$\text{plim}_{n \rightarrow \infty}(\hat{\tilde{\beta}}_1) = \beta_1$$

Apply the probability limit to the estimator, which yields the ratio of the population moments (due to LLN):

$$\text{plim}(\hat{\tilde{\beta}}_1) = \frac{\text{Cov}(X_1, Y)}{\text{Var}(X_1)}$$

Substitute the True Model for Y into the numerator, exploiting linearity and the standard OLS assumption $\text{Cov}(X_1, u) = 0$:

$$\begin{aligned} \text{plim}(\hat{\tilde{\beta}}_1) &= \frac{\text{Cov}(X_1, \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u)}{\text{Var}(X_1)} \\ \text{plim}(\hat{\tilde{\beta}}_1) &= \frac{0 + \beta_1 \text{Var}(X_1) + \beta_2 \text{Cov}(X_1, X_2) + 0}{\text{Var}(X_1)} \end{aligned}$$

This gives the general result for the probability limit, which includes the inconsistency term:

$$\text{plim}(\hat{\beta}_1) = \beta_1 + \beta_2 \left(\frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)} \right)$$

The inconsistency term vanishes because the problem explicitly states that X_1 and X_2 are independent in the population:

If X_1 and X_2 are independent, then $\text{Cov}(X_1, X_2) = 0$

Substitute:

$$\text{plim}(\hat{\beta}_1) = \beta_1 + \beta_2 \left(\frac{0}{\text{Var}(X_1)} \right)$$

$$\text{plim}(\hat{\beta}_1) = \beta_1$$

Since the probability limit of the estimator equals the true parameter, consistency is established.

it is a consistent estimator.

1.6 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a zip file for you to submit.
Please save before exporting!